What Can Forecasting Audits Establish?

Expert Testing and AI Auditing

Aaron Roth

University of Pennsylvania

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- Whatever process generates the incidents, a forecaster that reports the true probabilities should rarely fail the test.
- If we must give a verdict by a fixed date, a provider who knows only the test can randomize which forecasting rule it submits and pass with high probability on any fixed outcome sequence.
- Without a deadline, we can design a test that no strategy reliably passes on *every* outcome sequence. But the sequence that would expose a strategy need not be the one we see.

Calibration Does Not Certify Knowledge

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Beyond calibration

Calibration alone does not show that the forecaster knew the pattern. Could a different test establish that it knew how the outcomes were generated?

A Rule Gives a Forecast After Every History

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A *forecasting rule* f specifies

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For a law P on $\mathcal{Y} := \{0, 1\}^T$, the truthful forecast after a history h of length $t - 1$ is

$$f_P(h) = P(y_t = 1 \mid y_{<t} = h).$$

At histories with probability zero under P, the conditional probability is undefined; we set $f_P(h) = 0$ to specify a complete rule.

The Audit Should Rarely Reject Truthful Forecasts

The provider knows the test before choosing its forecasting rule f. The rule does not affect which incidents occur. After T days the auditor applies a deterministic test to the rule and the observed sequence y:

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Fix $0 < \alpha < 1$. If the outcome law is P and the forecaster reports its true probabilities f_P, the chance of rejection must be at most α. Equivalently, we require

$$\mathbb{P}_{y \sim P}\big(A(f_P, y) = 1\big) \geq 1 - \alpha \qquad \text{for every law } P \text{ on } \mathcal{Y}.$$

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This must hold even if the chance of an incident depends on earlier outcomes. The auditor may inspect f even at histories that did not occur.

The Provider Can Choose Its Forecaster by Coin Flip

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- Heads: submit the rule that reports 20% every day.
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An Uninformed Provider Can Pass Reliably

Theorem (Finite-horizon expert-testing impossibility)

If A rejects truthful forecasts with probability at most α under every outcome law, there is a distribution ζ over finitely many forecasting rules such that

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Fix any sequence of outcomes. Before seeing it, the provider draws a rule using only its knowledge of the test. With probability at least $1-\alpha$, that rule passes on this sequence.

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The same lottery works for every sequence; the same individual rule need not. Each submitted rule can still use past observations.

How Can the Provider Pass Without Knowing the Law?

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- The maximization player is the provider, choosing a rule f.
- The minimization player is Nature, choosing a sequence $y \in \mathcal{Y}$.
- The provider's payoff is $A(f, y)$: one for a pass and zero for a rejection. Its expected payoff is therefore the probability of passing.

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There are $|\mathcal{Y}| = 2^T$ possible sequences. Each row is a list of $|\mathcal{Y}|$ zeros and ones:

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Only finitely many such rows exist. Write $\mathcal{F}_0$ for the rules we kept, one per distinct row. Removing duplicate rows changes no payoff, so the finite minimax theorem applies.

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The representative of f_P in $\mathcal{F}_0$ gives the same verdicts. Thus

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This is a response to each *known* law. Minimax will give one randomized strategy that works without knowing the law.

Minimax Removes the Need to Know the Outcome Law

For a finite set S, write $\Delta(S)$ for its probability distributions. The previous slide gives

$$\min_{P \in \Delta(\mathcal{Y})} \max_{f \in \mathcal{F}_0} \mathbb{E}_{y \sim P}[A(f, y)] \geq 1 - \alpha.$$

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$$\max_{\zeta \in \Delta(\mathcal{F}_0)} \min_{y \in \mathcal{Y}} \mathbb{E}_{f \sim \zeta}[A(f, y)].$$

Choose a maximizing ζ. Then, for *every* fixed sequence y,

$$\mathbb{P}_{f \sim \zeta}\big(A(f, y) = 1\big) = \mathbb{E}_{f \sim \zeta}[A(f, y)] \geq 1 - \alpha.$$

This proves the theorem. □

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So far the auditor had to give a verdict by day T. What if it could keep watching and reject only when it sees evidence against the forecast?

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The auditor could already inspect forecasts for histories it never saw. What is new is that it can reject on *any* finite day and need never announce acceptance.

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The test in words

Find a day n such that, according to the model, the chance that the first incident occurs *after* day n, but still occurs, is at most 5%. Reject if that is what happens.

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A long incident-free run is not grounds for rejection. The auditor waits for an actual incident. Why can it always find such a cutoff n?

Every Announced Law Has a Finite Cutoff

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Under the announced law Q, the probabilities $Q(\tau = t)$ of all *finite* first-incident days sum to at most one. The tails of that series tend to zero:

$$Q(n < \tau < \infty) = \sum_{t=n+1}^{\infty} Q(\tau = t) \longrightarrow 0 \qquad (n \to \infty).$$

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The auditor can therefore choose the first cutoff that works:

$$n(Q) := \min\{n \geq 0 : Q(n < \tau < \infty) \leq \alpha\}.$$

It rejects Q exactly when $n(Q) < \tau < \infty$.

Every Reporting Strategy Fails on Some Sequence

Theorem (The late-incident test)

1. *For every true law P, a truthful report $Q = P$ satisfies*

$$\mathbb{P}_{y \sim P}(\textit{reject } P) \leq \alpha.$$

2. *For every distribution ζ over reports and every $\delta > 0$, there is a fixed sequence y on which*

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Part 1 follows from how we chose the cutoff: if $Q = P$, the probability of rejection is $P(n(P) < \tau < \infty) \leq \alpha$.

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Part 1 follows from how we chose the cutoff: if $Q = P$, the probability of rejection is $P(n(P) < \tau < \infty) \leq \alpha$.
Part 2 is called *nonmanipulability*: for each lottery ζ, we can find a sequence y that makes rejection likely without knowing which report Q was drawn.

Choose a Day Later Than Almost All the Cutoffs

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Why can we choose such a day? Every draw $Q \sim \zeta$ gives a finite cutoff. Thus for any $\delta > 0$ we can choose a finite t with

$$\mathbb{P}_{Q \sim \zeta}\big(n(Q) < t\big) \geq 1 - \delta,$$

because the probability on the left tends to one as $t \to \infty$.

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Fix the sequence with its first incident on day t and none thereafter. On this sequence, $\tau = t$, so

$$\mathbb{P}_{Q\sim\zeta}(\text{reject } Q) = \mathbb{P}_{Q\sim\zeta}\big(n(Q) < t\big) \geq 1 - \delta. \qquad \square$$

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With a deadline, finite minimax turned a good response to each law into one strategy that worked against every sequence. Here no single lottery works: there is always a later day.

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No. The test rejects only when a first incident *occurs* after the cutoff. No incident means no rejection, however long the auditor waits.

What the theorem says

For each reporting strategy there is *some* sequence on which rejection is highly likely. It need not be likely under the true law P.

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Summary

Does passing a forecasting audit show that the AI knew the true probabilities in advance? *Not necessarily.*

- **With a deadline:** Suppose the test passes truthful forecasts at least 95% of the time, whatever process generates the outcomes. A provider who knows only the test can randomize which rule it submits and pass with probability at least 95% on *every* fixed outcome sequence.
- **Without a deadline:** The late-incident test still rarely rejects truthful forecasts. Yet for each reporting strategy, there is *some* sequence on which the test is very likely to reject it. That need not be the sequence we observe.

Thanks!

See you next class!

References

- Alvaro Sandroni. *The Reproducible Properties of Correct Forecasts*. *International Journal of Game Theory* 32, 151–159, 2003.
- Wojciech Olszewski and Alvaro Sandroni. *A Nonmanipulable Test*. *The Annals of Statistics* 37(2), 1013–1039, 2009.
- Lance Fortnow and Rakesh V. Vohra. *The Complexity of Forecast Testing*. *ACM SIGecom Exchanges* 7(3), 2008.

The finite-horizon proof uses finite verdict patterns and minimax. The positive test simplifies the Olszewski–Sandroni construction to the single reference path $0, 0, \ldots$, allows false rejection $\alpha > 0$, and proves near-certain rejection on some path rather than their stronger global category conclusion. Their Section 3.2 also distinguishes nonmanipulability from likely detection of an uninformed expert.